

PHYS 225

Week 7, Lecture 7:
Invariant Interval and Spacetime Metric



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reminders

- Homework 6 due at 11:59 pm this Thursday, 3/5.
- Homework 7 due **Monday 3/8!** It's a short one! Will be posted on Wednesday. Will also help you study for midterm.

- **Midterm is next week, Tuesday 3/10 at 4-5 pm**

- **Location:** know your Discussion Section!

Sections **D5A, B, E, F, J, K, M** – Loomis 141.

Sections **D5N, D5R, D5S** – LUMEB 2100.

- A practice exam is posted (see schedule page, Week 7).
- If you need DRES accommodations, please set them up ASAP!



- Don't forget to turn in your **1-minute papers** by 6 pm today!!!

Week 7, Lecture 7

Invariant Interval and Spacetime Metric

Objectives:

- Define the invariant interval and show its invariance under L.T.'s
- Classify spacetime displacements as timelike, spacelike, or lightlike
- Draw a Minkowski (spacetime) diagram and use it to solve conceptual problems
- Manipulate 4-vectors using the spacetime metric

One Minute Paper: draw a pair of coordinate axes to represent x and ct , and draw the path of a light ray on those axes.

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Invariant Interval

Lots of things change between reference frames: times, lengths, order of events, etc. It would be very nice to have a quantity that **doesn't** change — such a quantity would be **invariant**, and take the same values in all reference frames.

Here's an idea:

$$\Delta s^2 \equiv c^2(\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2$$

Sanity check: $\Delta s^2 = 0 \implies \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = c\Delta t$

This is indeed satisfied for any light beam in any inertial frame.

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Invariant Interval

The beauty of the invariant interval is that this works for **any** spacetime displacement, not just paths of light beams!

Synchronizing the coordinate systems:

$$\Delta s^2 = (x^0)^2 - (x^1)^2 - \underbrace{(x^2)^2 - (x^3)^2}$$

For Lorentz boosts along the x-axis,
 $y' = y$ and $z' = z$, so this part doesn't change

Goal: show that $(x'^0)^2 - (x'^1)^2 = (x^0)^2 - (x^1)^2$

[to blackboard]

Invariance of the interval:

$$ct' = \gamma ct - \gamma \beta x$$

$$x' = -\gamma \beta ct + \gamma x$$

$$\begin{aligned} (ct')^2 - (x')^2 &= (\gamma ct - \gamma \beta x)^2 - (-\gamma \beta ct + \gamma x)^2 = \\ &= \underbrace{\gamma^2 c^2 t^2} + \underbrace{\gamma^2 \beta^2 x^2} - 2\gamma^2 \beta ct x - \\ &\quad - \underbrace{\gamma^2 \beta^2 c^2 t^2} - \underbrace{\gamma^2 x^2} + 2\gamma^2 \beta ct x = \\ &= \gamma^2 (1 - \beta^2) c^2 t^2 - \gamma^2 (1 - \beta^2) x^2 = \\ &= \gamma^2 (1 - \beta^2) (c^2 t^2 - x^2) \end{aligned}$$

$$\gamma^2 = \frac{1}{(1 - \beta^2)}$$

$$\Rightarrow \gamma^2 (1 - \beta^2) = 1 \Rightarrow \boxed{(ct')^2 - (x')^2 = (ct)^2 - x^2}$$

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Rotations and Lorentz transformations

As you will show in Discussion 7, this also works for boosts along different axes. There is a very close analogy to rotations:

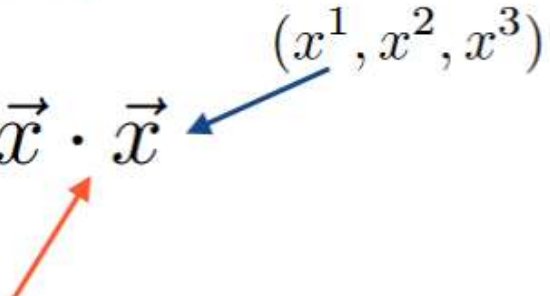
Coordinate transformation	Matrix representation	Invariant quantity
Rotations	Orthogonal matrices	Dot product
Lorentz transformations	Lorentz matrices	Invariant interval

In fact, rotations and Lorentz boosts can be combined into a single group, the **Lorentz group** (Discussion 7)

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Rotations and Lorentz transformations

Let's make the connection even more explicit:

$$\Delta s^2 = (x^0)^2 - \vec{x} \cdot \vec{x}$$


Dot product is contained
in invariant interval!

3-dimensional rotations preserve the dot product
(Discussion 6) and don't touch the time coordinate.
Therefore, rotations **also** preserve the invariant interval!

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Classification of invariant intervals

Since the invariant interval doesn't change between inertial frames, we can use it to classify the spacetime displacement between two events. The first event will be at (0,0) in all frames.

$\Delta s^2 > 0$: **timelike** separation

$$c^2 t^2 > x^2 \implies |x/t| < c$$

After a Lorentz transformation: $x' = \gamma(x - vt)$

Pick $v = x/t$: this is a valid frame, in which $x' = 0$. In this frame, events occur at the same place, but separated in **time**:

$\Delta\tau = t' = \sqrt{\Delta s^2}/c$ **invariant interval measures proper time: time measured by an observer who travels between events**

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Classification of invariant intervals

Since the invariant interval doesn't change between inertial frames, we can use it to classify the spacetime displacement between two events. The first event will be at (0,0) in all frames.

$$\Delta s^2 < 0 : \quad \text{spacelike separation}$$

$$c^2 t^2 < x^2 \implies |x/t| > c$$

No valid L.T. can bring these events to the same place! However, we can make them occur at the same time, but separated in **space**:

$$t' = \gamma(t - vx/c^2) \implies \text{choose } v = c^2 t/x \text{ to get } t' = 0$$
$$\Delta L = x' = \sqrt{-\Delta s^2} \quad \text{invariant interval measures proper length}$$

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Spacelike intervals and causality

Since the invariant interval doesn't change between inertial frames, we can use it to classify the spacetime displacement between two events. The first event will be at (0,0) in all frames.

$$\Delta s^2 < 0 : \quad \text{spacelike separation}$$

Spacelike-separated events can never affect one another: if we think event A **caused** event B, we could always transform to a frame where event B happens **before** event A: contradiction!

$$t' = \gamma(t - vx/c^2)$$
$$v > \frac{c^2 t}{x} \implies t' < 0$$

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Classification of invariant intervals

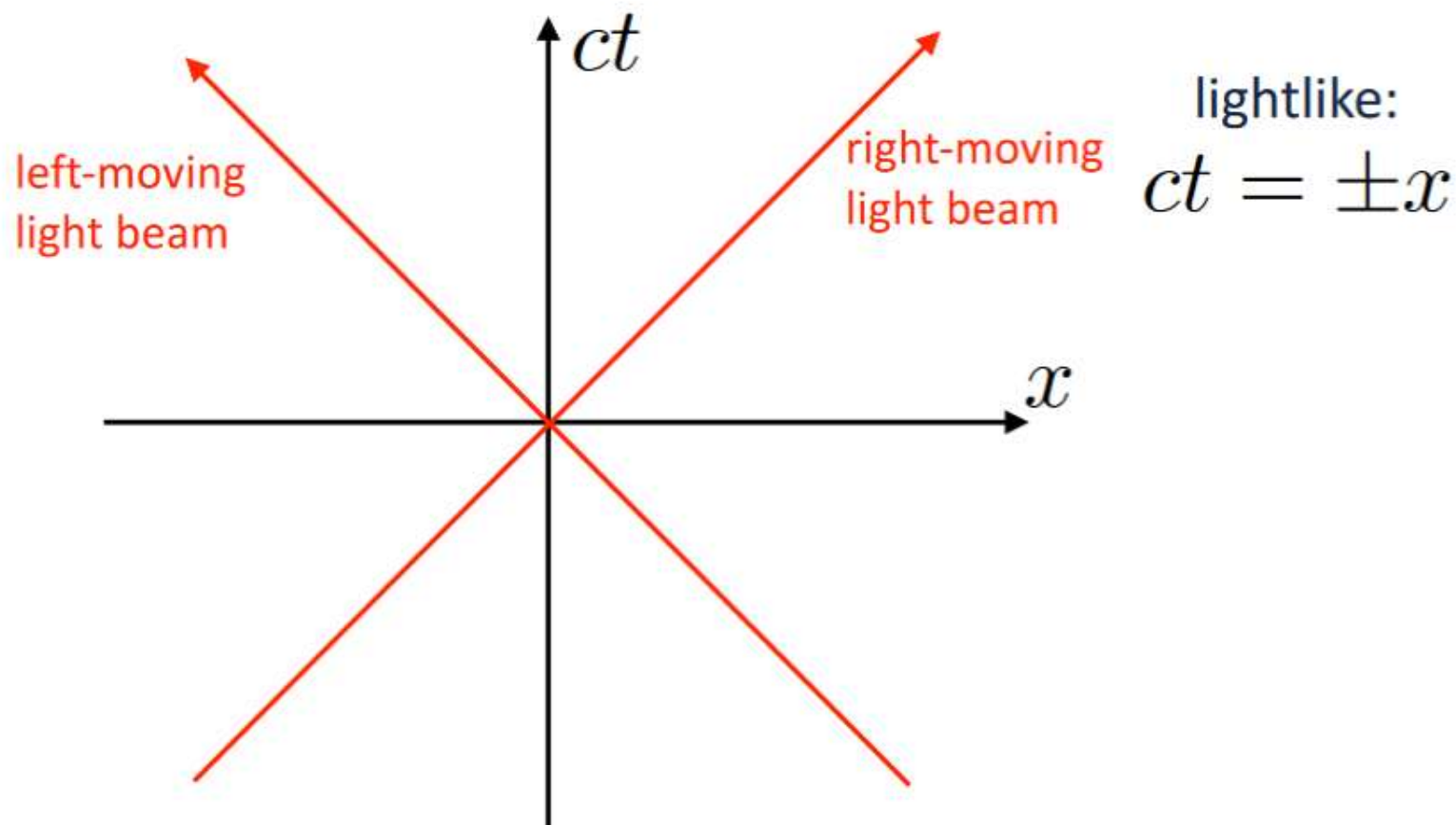
Since the invariant interval doesn't change between inertial frames, we can use it to classify the spacetime displacement between two events. The first event will be at (0,0) in all frames.

$$\Delta s^2 = 0 : \quad \text{lightlike separation}$$

Events for which the invariant interval vanishes are connected by paths of light beams. No notion of proper time or proper length: constancy of interval just means light has speed c in every inertial frame.

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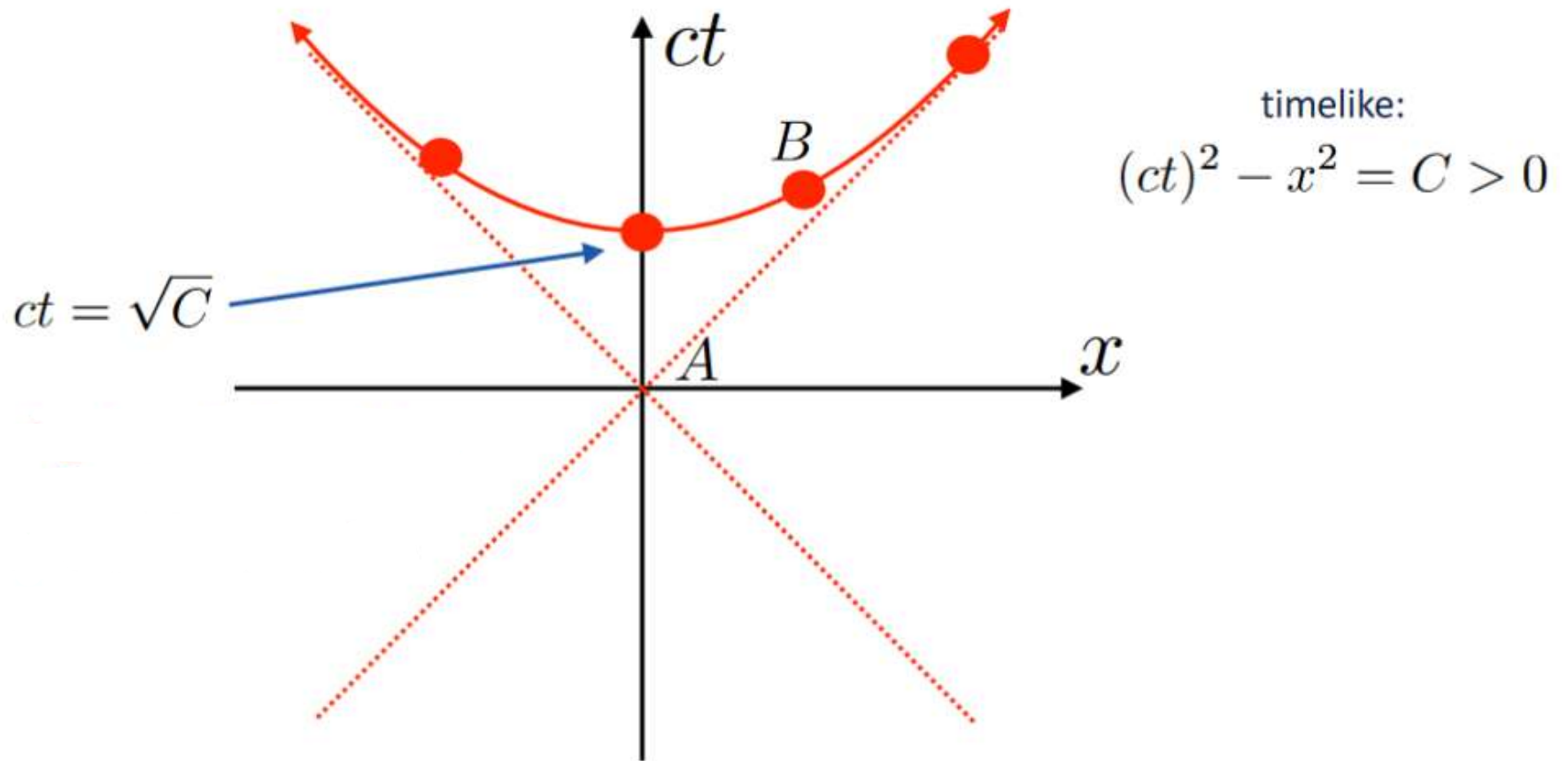
Minkowski (spacetime) diagrams



On x and ct axes, paths of light beams are straight lines at 45°

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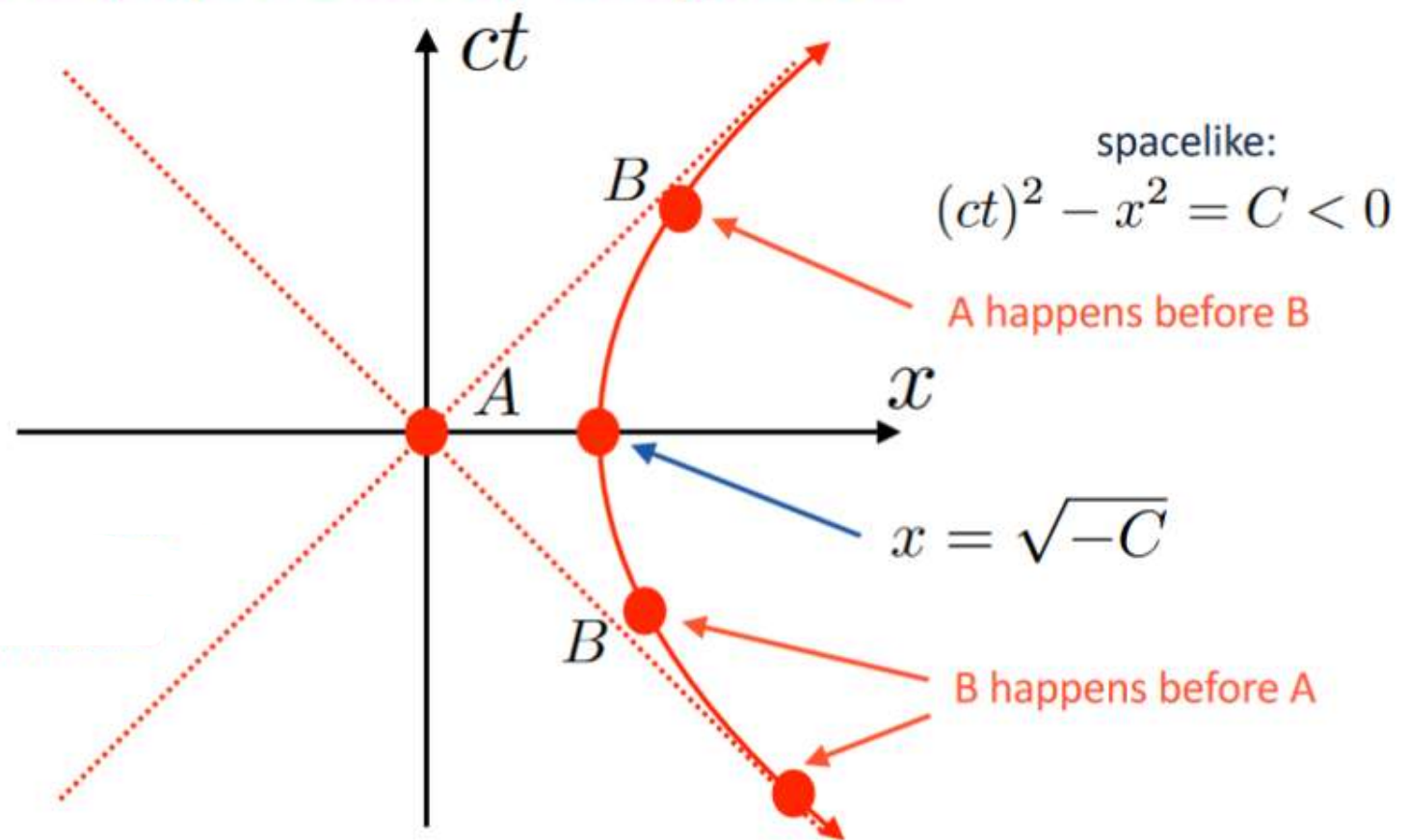
Minkowski (spacetime) diagrams



Locus of timelike-separated events is a vertical hyperbola w/asymptotes at 45°

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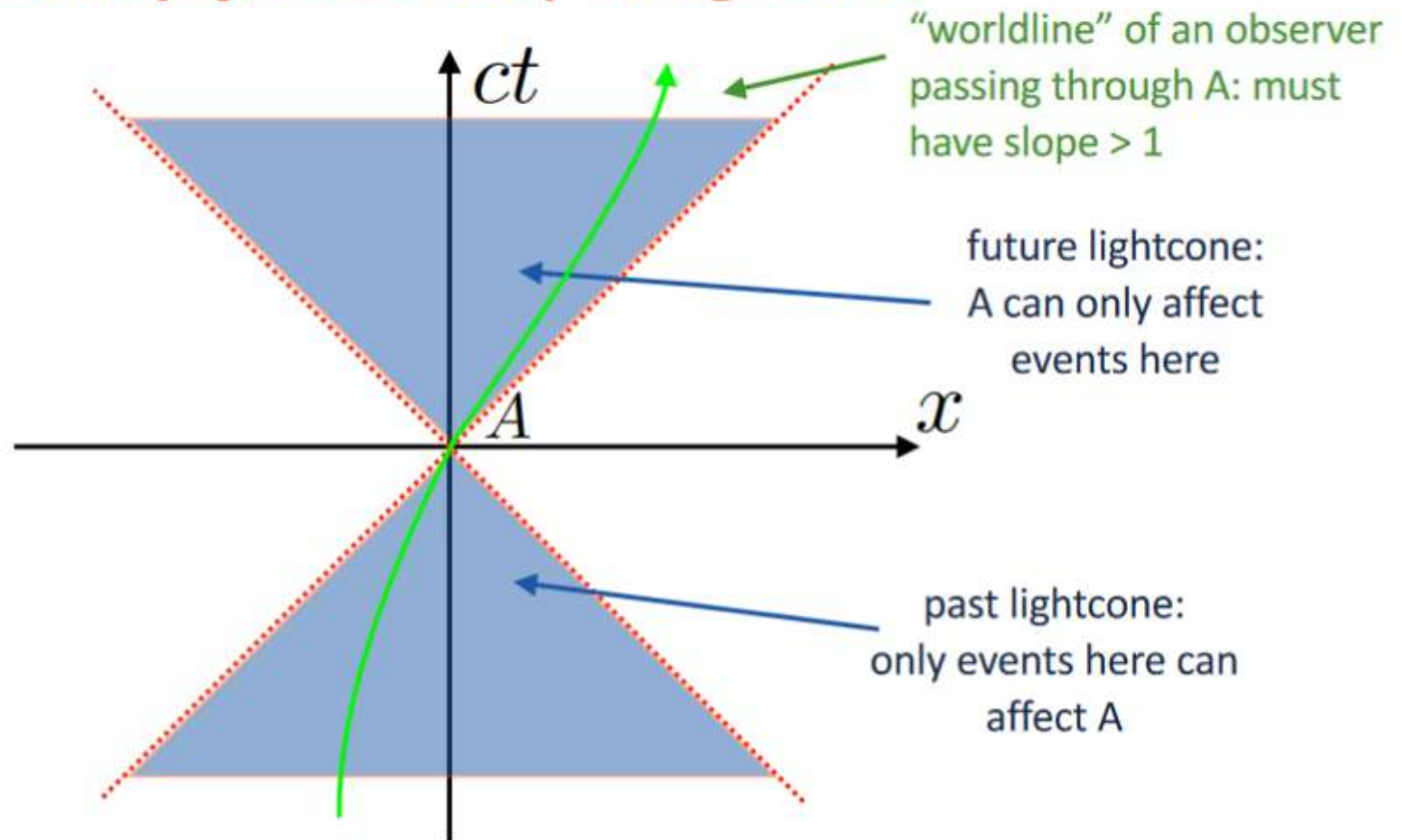
Minkowski (spacetime) diagrams



Locus of spacelike-separated events is a horizontal hyperbola w/asymptotes at 45°

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Minkowski (spacetime) diagrams

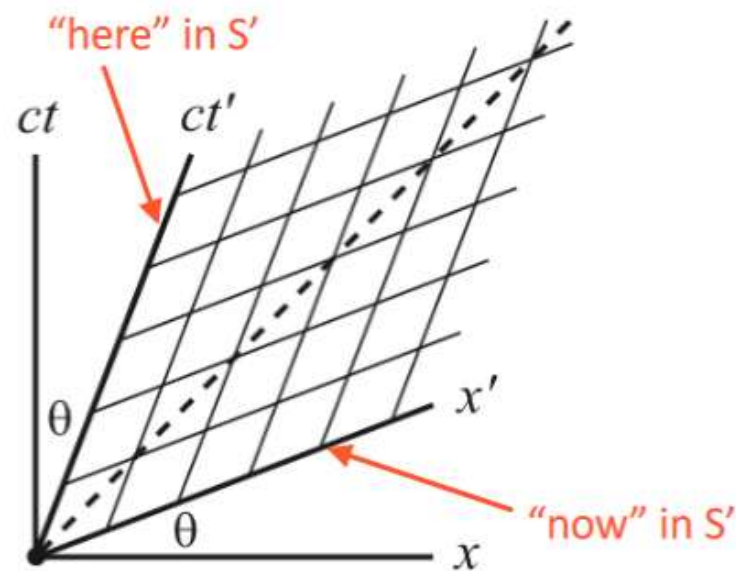
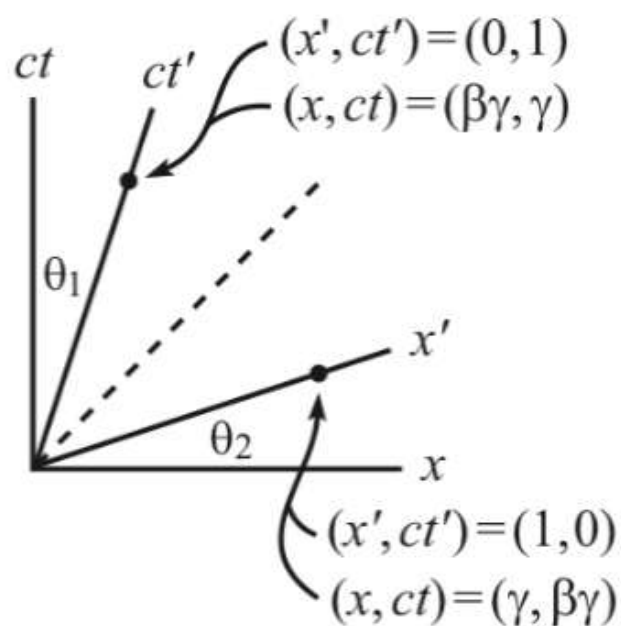


(Have to imagine y and z coordinates at each point)

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Minkowski (spacetime) diagrams

If we want to represent **two** frames S and S' on the same diagram:

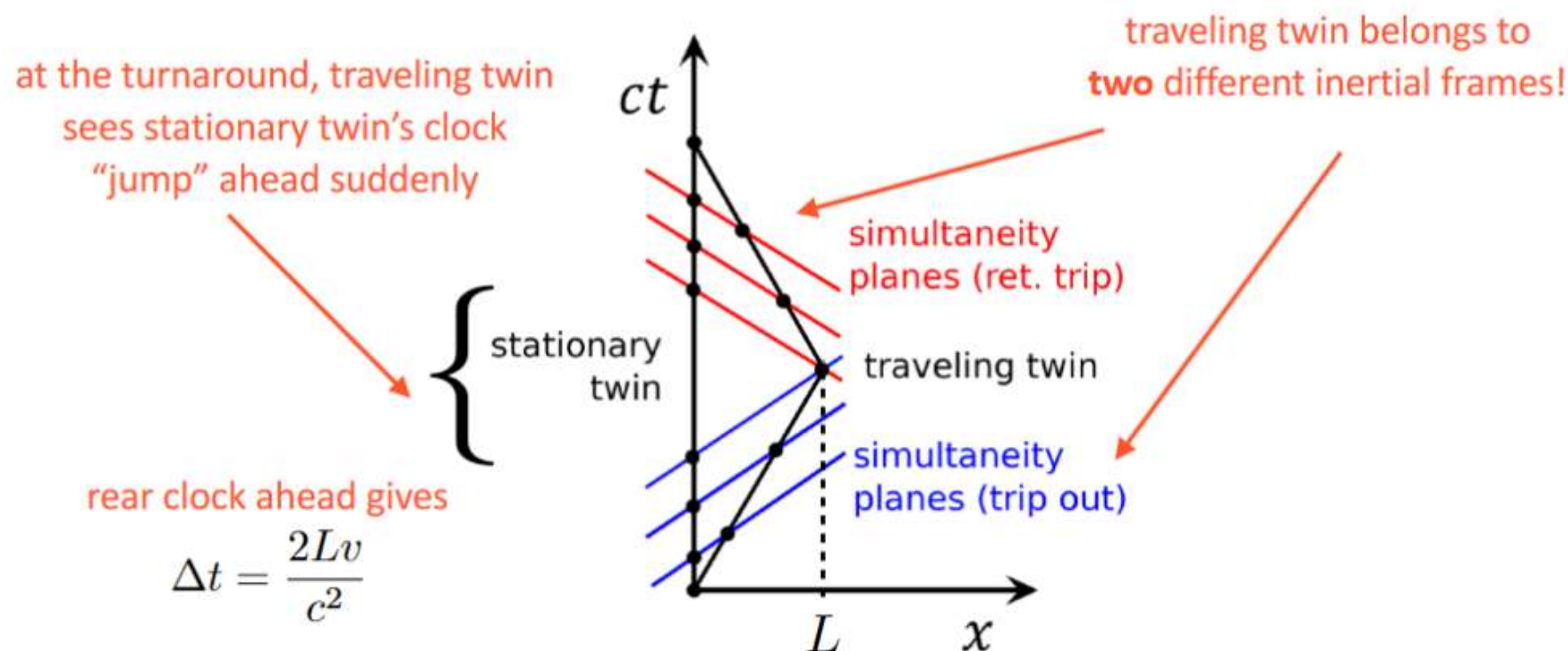


$$\tan \theta_1 = \tan \theta_2 \equiv \tan \theta = \beta \quad (\text{Morin 2.4 and Discussion 7})$$

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Minkowski (spacetime) diagrams

Minkowski diagrams help us understand the twin paradox:



Very useful for qualitative understanding; Lorentz transformations better for quantitative

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The spacetime metric

The rest of this lecture will **not** appear on the midterm. But it is crucial for the remainder of the course (and the final!), so pay close attention!

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The spacetime metric

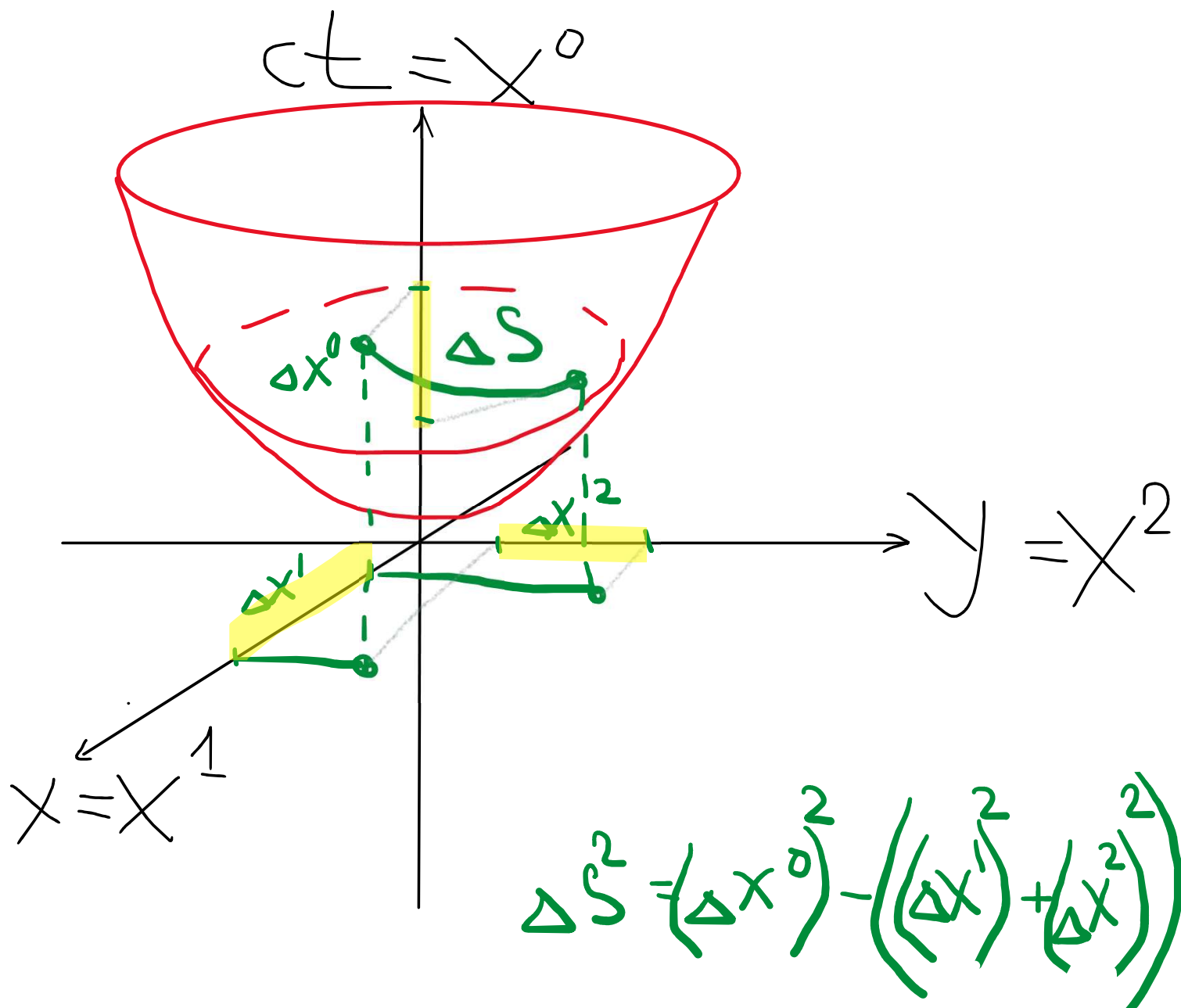
Recall from lectures 3 and 6: $\vec{x} \cdot \vec{y} \equiv \delta_{ij} x^i y^j$

Including the time coordinate, we can write the invariant interval as:

$$\Delta s^2 = g_{\mu\nu} (\Delta x^\mu) (\Delta x^\nu)$$

$$g_{\mu\nu} = \begin{cases} 1 & \mu = \nu = 0 \\ -1 & \mu = \nu = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

$g_{\mu\nu}$ is called the **spacetime metric**.



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The spacetime metric

The metric lets us lower indices:

$$x^\mu = (x^0, x^1, x^2, x^3)$$

$$x_\mu = g_{\mu\nu}x^\nu = (x^0, -x^1, -x^2, -x^3) = (x^0, -\vec{x})$$

$$x_0 = x^0, \quad x_1 = -x^1, \quad x_2 = -x^2, \quad x_3 = -x^3$$

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The spacetime metric

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$$x_\mu = g_{\mu\nu}x^\nu = (x^0, -x^1, -x^2, -x^3) = (x^0, -\vec{x})$$

$$x_0 = x^0, \quad x_1 = -x^1, \quad x_2 = -x^2, \quad x_3 = -x^3$$

WARNING!!! There are different sign conventions for the metric. Our convention in this course is called the “mostly minus” convention. Always check!

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The spacetime metric

We can also raise indices using the inverse metric (which has the same numerical values as the metric itself):

$$g^{\mu\nu} = \begin{cases} 1 & \mu = \nu = 0 \\ -1 & \mu = \nu = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

$$x^\mu = x_\nu g^{\mu\nu} = g^{\mu\nu} x_\nu$$

Spacetime vectors with upstairs indices are like column vectors, those with downstairs indices are like row vectors

$$g_{\mu\sigma} g^{\sigma\nu} = \delta^\nu_\mu \iff gg^{-1} = I$$

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Lorentz invariance

Define a compact notation which we will use extensively after break:

$$x \cdot y \equiv g_{\mu\nu} x^\mu y^\nu = x^\mu y_\mu = x_\mu y^\mu$$

This is called the **relativistic dot product**.

In particular, $\Delta s^2 = \Delta x \cdot \Delta x = \Delta x_\mu \Delta x^\mu$

We should require that the dot product be **Lorentz invariant** (takes same value after Lorentz transformation) for boosts along any axis:

$$\begin{aligned} x'^\mu &= \Lambda^\mu_\rho x^\rho \\ x \cdot x &= g_{\rho\sigma} x^\rho x^\sigma \\ x' \cdot x' &= g_{\mu\nu} x'^\mu x'^\nu = g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma x^\rho x^\sigma \end{aligned} \quad \Longrightarrow \quad g_{\rho\sigma} = g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma$$

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Invariant Interval and Spacetime Metric

Summary

- Invariant interval takes the same value in all inertial frames
- Two events can be timelike, spacelike, or lightlike separated, and this classification holds in all inertial frames
- Only timelike or lightlike separated events can be in causal contact
- Locus of an event on a single set of spacetime axes is a hyperbola; different frames have tilted axes
- Spacetime metric is analogous to Kronecker delta for dot products, but watch the minus signs!

One Minute Paper: draw a Minkowski diagram with S and S' axes when the boost velocity of S' approaches the speed of light